A LEARNING GUIDE FOR HOTEL REVENUE ANALYTICS

How Hotel Demand Builds

A Step-by-Step Guide to Booking Pace and Pickup Analysis

Learn how to read lead times, pace curves, heatmaps and booking blocks, using a real analysis of 75,166 stayed bookings from a City Hotel and a Resort Hotel.

Who this guide is for

This guide is for students, analysts and hotel professionals who want to understand how booking pace and pickup analysis works, from the first data-cleaning step to the final interpretation. No prior revenue management knowledge is needed. Every term is explained when it first appears, every method is shown with a small example, and every result is explained in plain language.

How to use it

Read Part 1 first: it defines the vocabulary used everywhere else. Part 2 walks through the ten steps of the analysis in order. Each step follows the same pattern: the question it answers, what was done, how to read the result, what the data showed, and why it matters for hotel decisions. Part 3 ties the steps together and ends with questions to check your understanding.

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3 Putting it together	The questions the analysis answers, how to read results carefully, and a self-check quiz with answers

ABOUT THE DATA

The Hotel Booking Demand dataset (Antonio, Almeida and Nunes, 2019) contains 119,390 bookings and 32 columns for a City Hotel and a Resort Hotel, with arrival dates from July 2015 to August 2017. All figures in this guide were calculated directly from that data. Reference: Antonio, N., Almeida, A. and Nunes, L. (2019). Hotel booking demand datasets. Data in Brief, volume 22.

PART 1

Foundations

The ideas and vocabulary behind pace and pickup analysis.

Why booking pace matters

A hotel sells a perishable product: a room that stays empty tonight can never be sold tomorrow. Revenue managers therefore need to know two things for every future arrival date: how many bookings are already confirmed, and how many more are likely to arrive before the date. Booking pace answers the first question. Pickup answers the second. Together they let a hotel judge whether a date is filling faster or slower than usual, and decide whether to raise prices, lower them, or restrict sales.

KEY IDEA

Pace is a snapshot of how full a date is at a given number of days before arrival. Pickup is the new business added between two snapshots.

Key terms

Term	Plain meaning	Example
Stayed booking	A booking where the guest actually checked out. Cancellations and no-shows are excluded.	75,166 of the 119,390 bookings in this dataset
Arrival date	The calendar date the guest checks in.	5 December 2015
Booking date	The date the reservation was made. The dataset does not store it, so it is derived (see Step 1).	11 July 2015
Lead time	The number of days between the booking date and the arrival date. It is also called the booking window.	30 days
Pace	How many bookings are on the books at a chosen number of days before arrival.	255 bookings on the books by arrival day
Pickup	Bookings added during a period, for example between 30 days and 7 days before arrival.	47 new bookings in the final week
Pace curve	A line that accumulates bookings as the arrival date gets closer, showing how a date fills up.	A line that climbs in steps and ends at the total
Cumulative share	The percentage of all bookings made within a given number of days of arrival.	41.9% of bookings are made within 30 days
Median and mean	The median is the middle value. The mean is the average. They differ when a few values are very large.	Median lead time 45 days, mean 80 days
ADR	Average daily rate: the average price per room per night recorded on a booking.	EUR 99.99 across all stayed bookings
Market segment	The type of customer or sales channel that produced the booking.	Online TA, Direct, Groups, Corporate
Block booking	A large number of bookings for the same arrival date made on a single day, typically by a group or a travel trade partner.	42 bookings made 11 days before one date
Heatmap	A grid where colour shows how many bookings fall in each cell. Darker means more.	Rows are arrival dates, columns are lead times

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Three small worked examples

WORKED EXAMPLE

Lead time. A guest books on 11 July for arrival on 10 August. Lead time = 10 August minus 11 July = 30 days. In reverse, if you know the arrival date (10 August) and the lead time (30 days), the booking date is 10 August minus 30 days = 11 July.

WORKED EXAMPLE

Median versus mean. Five bookings have lead times of 0, 2, 5, 10 and 200 days. The median is the middle value, 5 days. The mean is (0 + 2 + 5 + 10 + 200) / 5 = 43.4 days. One very early booking pulls the mean far above the median. Lead times behave this way in real data, which is why both are reported.

WORKED EXAMPLE

Cumulative share. If 100 bookings were made and 24 of them have a lead time of 7 days or less, then 24% of bookings were made within 7 days of arrival. In this dataset the figure is 23.7%.

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STEP 1

Preparing the data

How do we get from a raw booking list to a clean picture of when demand was made?

WHAT WAS DONE

- **Load** the bookings file: 119,390 rows and 32 columns, one row per booking.
- **Build the arrival date.** The file stores the arrival year, month and day in three columns. They are combined into one date so bookings can be grouped by arrival date.
- **Derive the booking date.** The file has no booking timestamp, so booking date = arrival date minus lead time.
- **Keep stayed bookings only.** Only bookings with status Check-Out are kept: 75,166 of 119,390. The 43,017 cancellations and 1,207 no-shows are removed.
- **Compute lead time in days** as arrival date minus booking date. It is the horizontal axis of every later view.

WHY IT IS DONE THIS WAY

Pace and pickup are about demand that became real business. A cancelled booking never produced a stay, so mixing it in would blend two different things. Deriving the booking date from lead time is the standard way to place a booking on the calendar when no timestamp exists.

KEY IDEA

Everything in this guide is measured relative to arrival. That is why lead time, not calendar date, is the main axis.

WATCH OUT

Because booking dates are derived rather than recorded, they are only as accurate as the lead time column. All results also describe honoured bookings only, not total booking requests.

WHAT THE DATA SHOWED

After filtering, 75,166 stayed bookings remain. They cover arrival dates from 1 July 2015 to 31 August 2017. Lead times range from 0 to 737 days.

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STEP 2

The booking window

How far ahead do guests book, and how is that spread over time?

WHAT WAS DONE

Two views of lead time were built. The first is a **histogram**: lead times are sorted into 50 equal-width groups (bins) and the number of bookings in each is counted. The second is a **cumulative curve**: for each number of days, it shows the share of all bookings made within that many days of arrival.

HOW TO READ IT

In the histogram, a tall bar on the left means many bookings were made very close to arrival, and a long thin stretch on the right means a few bookings were made far ahead. In the cumulative curve, a steep early rise means demand is concentrated near arrival. A curve that rises slowly means demand is spread across a long window.

WHAT THE DATA SHOWED

Measure	Result
Median lead time	45 days
Mean lead time	80 days
Booked on the arrival day	7.9% (5,915 bookings)
Within 7 days	23.7%
Within 30 days	41.9%
Within 60 days	56.3%
Within 90 days	66.4%
Within 180 days	85.9%
Within 365 days	98.6%
Longest lead time	737 days

Half of stayed bookings are made within 45 days of arrival, yet the average is 80 days. That gap is the signature of a right-skewed distribution: most bookings are fairly close to arrival, while a minority are made many months ahead and pull the average up.

WHY IT MATTERS

- It shows how far ahead the hotel can see its business. About 14% of bookings (those made more than 180 days ahead) give a long-range signal; the rest only appear in the final months.
- It sets sensible **forecast horizons**: how far ahead a forecast can lean on bookings already received.
- It frames **pricing windows**: almost one booking in four arrives in the last week, so late pricing decisions affect a large share of demand.

> **KEY IDEA**
> Use the median to describe the typical booking and the mean to remember that a long tail exists. Reporting only one of them hides half the story.

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STEP 3

The pace curve of one busy date

How does one date actually fill up, booking by booking?

WHAT WAS DONE

The busiest arrival date in the data, 5 December 2015 (255 stayed bookings), was isolated. Its bookings were sorted from the longest lead time to the shortest and counted cumulatively. The result is a **pace curve**: the number of bookings on the books at each point before arrival.

HOW TO READ IT

Imagine a line that starts on the far left, 135 days before arrival, at about 29 bookings, then climbs as the date approaches and ends at 255 on the arrival day. Flat stretches mean no new bookings. Vertical jumps mean many bookings arrived on the same day. A smooth slope means steady, individual bookings.

WHAT THE DATA SHOWED

- The curve moves in three large steps: **29 bookings made 135 days out** (all Groups, City Hotel), **29 made 47 days out** (28 of them Groups) and **42 made 11 days out** (41 of them Groups, split across both hotels).
- The median lead time for this date was 19 days, against 45 days for all bookings.
- At least 60 days before arrival, only **21.6%** of this date's final bookings were on the books. For all bookings the figure is **44.1%**. At least 30 days out the figures are **45.9%** and **58.6%**.

WORKED EXAMPLE

"On the books at least 30 days out" counts every booking whose lead time is 30 days or more. For 5 December 2015, 117 of its 255 bookings qualify, which is 45.9%. A typical booking in the dataset has a 58.6% chance of qualifying, so this date was built later than usual.

WHY IT MATTERS

A pace curve tells a revenue manager whether a future date is ahead of or behind the pace of comparable past dates. It also reveals when large blocks arrive, which is when pricing and inventory decisions matter most.

WATCH OUT

One date is one story. The curve is useful as a worked example and as a benchmark for similar dates, but it should not be read as the pace of every date.

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STEP 4

The pickup heatmap

Which arrival dates rely on last-minute pickup, and which are built early?

WHAT WAS DONE

All stayed bookings for arrivals in August 2017 (3,109 bookings) were arranged in a grid. Each row is one arrival date, each column is one lead time from 0 to 120 days, and each cell holds the number of bookings. The colour scale was capped at 10 so that ordinary cells stay readable while the densest ones still stand out. A bar for each date shows its total bookings across all lead times.

HOW TO READ IT

Look along a row to see when that date's bookings were made. A dark patch at the left edge means heavy pickup in the last days. A dark cell far to the right means a large early booking. A row that is pale on the left and speckled on the right was built early. Look down a column to see whether the same lead time is busy for many dates.

WHAT THE DATA SHOWED

- August 2017 averaged 100 bookings per arrival date, from 62 to 147.
- **11.2%** of the bookings were made within 3 days of arrival, **15.0%** within 7 days and **51.4%** within 120 days.
- Five cells reach 10 or more bookings: 24 August at 2 days out (14), 3 August at 2 days (12), 27 August on the day (12), 15 August on the day (11) and 20 August at 115 days out (10).
- The two cells at 2 days out are mostly Groups bookings (11 of 14 and 8 of 12). The cell at 115 days out is a single Offline TA/TO block of 10 bookings at the City Hotel.

WHY IT MATTERS

- It identifies **volatile dates** that depend on late pickup and **stable dates** that were built early.
- It supports decisions on **restrictions and pricing** close to arrival, such as minimum stays, for the dates that are still short of bookings.
- It shows that late pickup is not only individual guests: even in the final days, one group booking can create the densest cell of the month.

KEY IDEA

A heatmap trades detail for pattern. Use it to spot where the unusual cells are, then investigate those cells in the underlying data.

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STEP 5

City Hotel versus Resort Hotel

Do the two hotels have different booking behaviour?

WHAT WAS DONE

Stayed bookings were split by hotel. Each hotel's lead-time pattern was drawn in two ways: as **raw counts** (how many bookings at each lead time) and as a **cumulative share of the hotel's own bookings**. A summary table was then calculated for each hotel.

Why two views?

Raw counts mix timing and size. The City Hotel has 46,228 stayed bookings and the Resort Hotel has 28,938, so City sits higher at almost every lead time simply because it has more bookings. Expressing each hotel as a share of its own bookings removes size and leaves only timing.

> **KEY IDEA**
> To compare timing between groups of different size, compare shares, not counts.

WHAT THE DATA SHOWED

Measure	City Hotel	Resort Hotel
Stayed bookings	46,228	28,938
Median lead time	50 days	38 days
Mean lead time	80.7 days	78.8 days
Booked on the arrival day	6.1%	10.6%
Within 7 days	20.5%	28.9%
Within 30 days	39.3%	46.2%
Within 90 days	66.3%	66.6%
More than 180 days ahead	13.2%	15.6%
Average ADR	EUR 105.7	EUR 90.8
Average ADR, same-day bookings	EUR 94.9	EUR 71.6

The Resort Hotel books later: a higher share of its bookings arrive in the last week, and its median lead time is 12 days shorter. At 90 days the two hotels are almost identical, so the difference lies in the final weeks. The City Hotel earns a higher average rate.

> **WATCH OUT**
> A common assumption is that resorts book further ahead than city hotels. In this dataset the opposite holds for the median. Check assumptions against the data before building a strategy on them.

WHY IT MATTERS

Different hotels need different forecasts. A single set of pace benchmarks for both would overstate the early pace of the Resort Hotel and understate its late pickup.

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STEP 6

Weekend versus weekday arrivals

Does the day of the week change when guests book?

WHAT WAS DONE

Arrivals were classified by the weekday of the arrival date. **Weekend arrivals** are Friday, Saturday and Sunday. **Weekday arrivals** are Monday to Thursday. As in Step 5, both raw counts and cumulative shares were produced.

> **WATCH OUT**
> The two groups do not cover the same number of days: weekend has three and weekday has four. Weekday arrivals therefore have more bookings in raw counts (43,010 against 32,156) without being more popular per day.

WHAT THE DATA SHOWED

Measure	Weekend (Fri to Sun)	Weekday (Mon to Thu)
Stayed bookings	32,156	43,010
Median lead time	51 days	41 days
Mean lead time	82.2 days	78.3 days
Booked on the arrival day	5.8%	9.4%
Within 7 days	20.0%	26.5%
Within 30 days	39.0%	44.2%
More than 180 days ahead	14.3%	14.0%

Weekend arrivals are booked about ten days earlier at the median, and weekday arrivals have a larger share of last-minute bookings. Both groups have almost the same share of long-range bookings beyond 180 days.

WHY IT MATTERS

- It supports **day-of-week pricing**: weekday inventory depends more on late pickup, weekend inventory is committed earlier.
- It shapes **marketing timing**: weekend offers need to reach guests earlier than weekday offers.
- It improves **forecast accuracy**: a weekend date at 30 days out and a weekday date at 30 days out are at different stages of their typical build.

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STEP 7

Market segment pickup

Do different types of customer book on different timetables?

WHAT WAS DONE

Each booking belongs to a market segment. The five largest segments (98.9% of bookings) were compared with the same two views as before: raw counts by lead time and cumulative share of each segment's own bookings.

The segments

Segment	Who they are
Online TA	Guests who book through online travel agencies (websites and apps)
Offline TA/TO	Guests who book through traditional travel agents and tour operators
Direct	Guests who book directly with the hotel
Groups	Group bookings, often several rooms booked together
Corporate	Business accounts and company travel
Complementary	Complimentary stays
Aviation	Bookings linked to airlines and aviation

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WHAT THE DATA SHOWED

- **Online TA** is the largest segment (47.5% of bookings) and has both a heavy last-minute component and a long tail of early bookings.
- **Direct** and **Corporate** book late: 45.3% and 58.2% of their bookings arrive within a week, and 24.2% of Direct bookings are made on the arrival day.
- **Offline TA/TO** and **Groups** book early: 24.4% and 33.6% of their bookings are made more than 180 days ahead, with medians of 90 and 107 days.

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STEP 8

Market segment profile

What does each segment look like on one page?

WHAT WAS DONE

A profile table was built for every segment, combining size, timing and rate. A segment's share is its bookings divided by all stayed bookings.

WHAT THE DATA SHOWED

Segment	Bookings	Share	Median lead	Same-day	Within 7 days	Over 180 days	Avg ADR
Online TA	35,738	47.5%	43 days	5.5%	21.1%	9.7%	EUR 113.9
Offline TA/TO	15,908	21.2%	90 days	2.6%	11.2%	24.4%	EUR 83.4
Direct	10,672	14.2%	11 days	24.2%	45.3%	5.9%	EUR 113.9
Groups	7,714	10.3%	107 days	0.8%	7.3%	33.6%	EUR 77.1
Corporate	4,303	5.7%	6 days	15.6%	58.2%	0.6%	EUR 67.1
Complementary	646	0.9%	3 days	28.8%	70.0%	0.6%	EUR 3.1
Aviation	185	0.2%	3 days	17.8%	82.2%	0.0%	EUR 99.5

Median lead time separates the segments more clearly than any other measure, from 3 days for Complementary and Aviation to 107 days for Groups.

- Of the 17,844 bookings made within 7 days, 42.3% come from Online TA, 27.1% from Direct, 14.0% from Corporate and 10.0% from Offline TA/TO.
- Of the 10,615 bookings made more than 180 days ahead, 36.6% come from Offline TA/TO, 32.8% from Online TA and 24.4% from Groups.
- Online TA and Direct share the highest average rate (EUR 113.9). Groups (EUR 77.1) and Corporate (EUR 67.1) book at lower average rates.

WHY IT MATTERS

- It shows which segments **fill the base** early (Groups, Offline TA/TO) and which drive **late pickup** (Online TA, Direct, Corporate).
- It supports **channel mix** decisions: a segment with a high rate and a short lead time behaves very differently from one with a low rate and a long lead time.
- It is the right basis for **negotiations** with travel partners and corporate accounts.

KEY IDEA

A blended hotel-wide pace curve averages together segments whose median lead times range from 3 to 107 days. Segment-level curves describe demand build far more accurately.

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STEP 9

ADR by lead time

Do guests pay different rates depending on how far ahead they book?

WHAT WAS DONE

Two measures were created. The first is an **ADR-weighted pace curve**: for every lead time, the sum of ADR across all bookings. The second groups lead times into buckets (same day, 1 to 7, 8 to 30, 31 to 60, 61 to 90, 91 to 180, 181 to 365 and 366+ days) and calculates the **average ADR** in each bucket, overall and by hotel.

WATCH OUT

A sum of ADR rises with the number of bookings, so the ADR-weighted curve has almost the same shape as the booking count curve. It shows where revenue is concentrated, not where rates are high. Only the average ADR per bucket answers whether late or early bookings pay more.

WHAT THE DATA SHOWED

Lead time	Bookings	Share of bookings	Share of total ADR	Average ADR
Same day	5,915	7.9%	6.5%	EUR 82.8
1 to 7 days	11,929	15.9%	14.7%	EUR 92.4
8 to 30 days	13,677	18.2%	19.0%	EUR 104.6
31 to 90 days	18,412	24.5%	25.5%	EUR 104.0
91 to 180 days	14,618	19.4%	20.6%	EUR 106.2
181+ days	10,615	14.1%	13.7%	EUR 96.8

Same-day bookings carry the lowest average rate (EUR 82.8). Rates rise through the final month and are highest for bookings made 8 to 180 days ahead (EUR 104.0 to EUR 106.2), then fall again for the longest lead times (EUR 96.8). In both hotels, same-day bookings have the lowest average rate of any bucket: EUR 94.9 at the City Hotel and EUR 71.6 at the Resort Hotel.

WHY IT MATTERS

- It identifies the **most valuable part of the booking window** for rate: roughly one to six months before arrival.
- It informs **dynamic pricing**: late inventory should be priced knowing that same-day bookings historically pay about EUR 20 less than mid-window bookings.
- It warns against **over-discounting** windows where guests already pay more.

WATCH OUT

These are patterns, not causes. Rates also change with season, room type and segment, so a lower same-day rate does not prove that late booking itself lowers the price.

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STEP 10

The ten busiest arrival dates

What makes a date reach its peak?

WHAT WAS DONE

The ten arrival dates with the most stayed bookings were selected (181 to 255 bookings each). For each date, bookings were counted by lead time, and the single largest lead-time spike was traced to its market segment, hotel and share of the date's bookings.

HOW TO READ IT

A large spike at one lead time means many bookings were made for that date on the same day. That is the footprint of a block booking, such as a group or a travel trade allocation.

WHAT THE DATA SHOWED

Arrival date	Bookings	Largest block	Segment	Made	Share of date
5 Dec 2015	255	42	Groups	11 days out	16%
24 Jun 2016	220	95	Offline TA/TO	192 days out	43%
26 May 2016	213	122	Groups	316 days out	57%
6 Jun 2016	200	79	Offline TA/TO	118 days out	40%
25 May 2017	190	45	Groups	203 days out	24%
25 Feb 2017	189	71	Groups	134 days out	38%
15 Oct 2015	188	82	Groups	287 days out	44%
24 Mar 2016	184	31	Offline TA/TO	136 days out	17%
2 Oct 2015	183	90	Offline TA/TO	164 days out	49%
2 May 2016	181	26	Groups	199 days out	14%

- Every one of the ten largest blocks is a Groups or Offline TA/TO booking: six Groups and four Offline TA/TO.
- Blocks range from 26 to 122 bookings and were made between 11 and 316 days before arrival.
- On six of the ten dates, a single block makes up 38% to 57% of the date's bookings. Together the ten blocks account for 683 of the 2,203 bookings on these dates (31%).

WHY IT MATTERS

The busiest dates are not produced by an unusually strong stream of individual guests. They are produced by large contracted blocks placed on a single day. A forecast that treats all bookings alike will miss this. Tracking block pickup separately from individual pickup explains most of the variation between peak dates.

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PART 3

Putting it together

The questions this analysis answers and how to interpret the answers with care.

The questions revenue managers ask

Question	Where to look	Short answer from this data
When does demand appear?	Steps 2 to 4	Median 45 days before arrival; 23.7% within a week; 14.1% more than 180 days ahead
Which dates are risky and which are stable?	Steps 3, 4 and 10	Dates built from late pickup are volatile; dates built from early blocks are stable but depend on those blocks
Which segments and channels matter?	Steps 7 and 8	Segments run from a 3-day to a 107-day median lead time, with very different rates
How does price behave across the window?	Step 9	Lowest on the arrival day, highest 8 to 180 days out
Do hotels and weekdays behave differently?	Steps 5 and 6	Resort books later than City; weekend arrivals book earlier than weekday arrivals

Reading the results carefully

- **Stayed bookings only.** Cancelled bookings are excluded, so the results describe business that was honoured, not every booking request.
- **Derived booking dates.** Booking dates come from lead time, not from a booking system.
- **Two hotels, two years.** The data covers one City Hotel and one Resort Hotel from July 2015 to August 2017, so patterns reflect those properties and that period.
- **Booking records, not rooms.** Counts are bookings. The dataset has no room capacity, so it cannot show how full a hotel was.
- **Patterns, not causes.** A lower rate for late bookings, or a later booking window at the Resort, are observed patterns. They do not by themselves explain why.

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Check your understanding

1. A guest books on 3 March for arrival on 20 March. What is the lead time?

Answer: 17 days.

2. Why is the mean lead time (80 days) higher than the median (45 days)?

Answer: A minority of bookings are made many months ahead, and these long lead times pull the average up while leaving the middle value unchanged.

3. Why compare City and Resort by share of their own bookings and not by raw counts?

Answer: City has more bookings overall, so raw counts mostly show size. Shares remove size and show timing.

4. What does "45.9% on the books at least 30 days out" mean for 5 December 2015?

Answer: 117 of the date's 255 stayed bookings were made 30 or more days before arrival.

5. Why does a sum of ADR by lead time not show which bookings pay more?

Answer: It grows with the number of bookings. Only the average ADR per lead-time bucket compares rates.

6. What produced the peaks on the ten busiest dates?

Answer: Single large Groups or Offline TA/TO blocks of 26 to 122 bookings, made 11 to 316 days before arrival.

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